

Customer Segmentation & Churn Pattern Analytics in European Banking:

Design, Implementation, and Empirical Findings of an Interactive Streamlit Dashboard

A Data Analytics Case Study

Project repository: [european-bank-churn-dashboard](#)

[github.com/Coding-With-SouRav/european-bank-churn-dashboard](#)

Dataset: European_Bank.csv | 10,000 customer records | 3 countries

Technology stack: Python, Streamlit, Pandas, NumPy, Plotly

ABSTRACT

Customer churn – the loss of clients to competitors or account closure – is one of the most consequential problems in retail banking, where acquisition costs routinely exceed retention costs by a wide margin. This paper presents the design, architecture, and analytical findings of a modular, interactive churn-analytics dashboard built for a synthetic European banking dataset of 10,000 customers spanning France, Germany, and Spain. The system is implemented in Python using the Streamlit framework for the user interface, Pandas and NumPy for data preparation and feature engineering, and Plotly for interactive visualization. The application is organized into a clean, testable package structure that separates configuration, data loading, filtering logic, KPI computation, and four analytical views (Overall Summary, Geography, Age & Tenure, and a High-Value Customer Explorer). Beyond documenting the software architecture, we analyze the underlying dataset directly and report the empirical churn patterns it reveals: an overall churn rate of 20.4%, a churn rate in Germany (32.4%) roughly double that of France and Spain, a churn rate among customers with three or more products that exceeds 80%, and a pronounced peak in churn risk among customers aged 46-60. We discuss the implications of these findings for retention strategy, evaluate the strengths and limitations of the dashboard's design, and outline directions for future work, including the integration of predictive modeling and cohort-based retention analysis.

Keywords: *customer churn, banking analytics, Streamlit, data visualization, customer segmentation, exploratory data analysis, retention strategy, Plotly*

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1. Introduction

Customer attrition, commonly referred to as *churn*, is a central concern for retail banks operating in competitive, multi-country markets. Industry research consistently finds that acquiring a new customer costs several times more than retaining an existing one, which makes early identification of at-risk segments a high-leverage activity for customer relationship management (CRM) and marketing teams. However, churn is rarely uniform: it varies by geography, demographic profile, product ownership, engagement level, and account characteristics. Aggregate churn statistics can mask meaningfully different risk profiles across these dimensions, which is why banks increasingly rely on interactive analytics tools rather than static quarterly reports.

This paper documents a self-contained churn-analytics application built around a 10,000-row customer dataset from a European retail bank. The project, distributed as the open-source repository **european-bank-churn-dashboard**, implements an interactive Streamlit dashboard that lets analysts filter the customer base along six dimensions, monitor five headline KPIs in real time, and explore churn patterns across four dedicated analytical tabs. Our contribution is twofold. First, we describe the software architecture of the dashboard — a modular Python package that cleanly separates data ingestion, feature engineering, filtering, metric computation, and presentation, making the codebase easy to extend and maintain. Second, we go beyond the tool itself and report the substantive churn patterns present in the underlying data, computed directly from the source CSV file, so that the analytical claims in this paper can be independently reproduced.

The remainder of this paper is organized as follows. Section 2 situates the project within the broader context of churn analytics and BI-tool design. Section 3 describes the dataset and the derived features used throughout the dashboard. Section 4 details the system architecture and implementation. Section 5 walks through the dashboard's user-facing design. Section 6 presents the empirical churn findings, illustrated with figures reproduced from the dataset. Section 7 discusses the practical implications of these findings, Section 8 addresses limitations, and Section 9 concludes with directions for future work.

2. Related Work and Background

Churn prediction and analysis has a long history in both the data-mining and CRM literature. Classical approaches frame churn as a binary classification problem and apply logistic regression, decision trees, or ensemble methods (e.g., random forests and gradient boosting) to a labeled dataset of past customer behavior. A large body of applied work in the telecommunications and banking sectors has shown that tenure, product engagement, transaction activity, and demographic variables such as age are consistently among the strongest predictors of attrition.

Separate from predictive modeling, a complementary line of work focuses on *descriptive and diagnostic analytics*: building interactive tools that let business users explore churn drivers without requiring a trained model. Business-intelligence platforms such as Tableau and Power BI are the traditional vehicles for this kind of analysis, but the rise of Python-native app frameworks such as Streamlit and Dash has made it practical for data analysts and data scientists to build bespoke, code-first dashboards that are version-controlled, reproducible, and easy to deploy alongside the rest of a data pipeline. This project sits in that second tradition: rather than training a predictive model, it exposes the descriptive structure of churn in the customer base through segmentation, filtering, and comparative visualization, which is typically the first

step analysts take before any predictive modeling effort.

3. Dataset Description

The dashboard operates on **European_Bank.csv**, a structured dataset of 10,000 customer records with no missing values across any of its 14 original columns. Each row represents one bank customer and includes demographic attributes, account characteristics, and a binary churn label (*Exited*). Table 1 summarizes the raw fields.

Field	Type	Description
CustomerId	Integer	Unique customer identifier
Surname	Text	Customer surname (dropped during preprocessing for privacy)
CreditScore	Integer	Credit score (range observed: approx. 350-850)
Geography	Categorical	Country of residence: France, Germany, or Spain
Gender	Categorical	Male or Female
Age	Integer	Customer age in years (18-92)
Tenure	Integer	Years as a bank customer (0-10)
Balance	Float	Account balance (EUR); 36.2% of customers hold a zero balance
NumOfProducts	Integer	Number of bank products held (1-4)
HasCrCard	Binary	Whether the customer holds a credit card (70.6% do)
IsActiveMember	Binary	Whether the customer is an actively engaged member
EstimatedSalary	Float	Estimated annual salary (EUR)
Exited	Binary	Target label: 1 = churned, 0 = retained

Table 1. Raw fields present in *European_Bank.csv* ($n = 10,000$; 0 missing values).

3.1 Feature Engineering

The data-loading layer (`dashboard/data.py`) performs several derivations before any record reaches the interface, so that every downstream chart and filter operates on consistent, pre-binned categories:

- **AgeGroup** — Age is binned into four cohorts: <30, 30-45, 46-60, and 60+.
- **CreditScoreBand** — Credit score is binned into Low (≤ 579), Medium (580-699), and High (> 699), following common industry thresholds for subprime, near-prime, and prime borrowers.
- **TenureGroup** — Tenure is grouped into New (0-2 years), Mid-term (3-6 years), and Long-term (7+ years) relationships.
- **BalanceSegment** — Customers are split into Zero-balance, Low-balance, and High-balance groups, with the Low/High threshold set at the median balance among customers with a strictly positive balance (€119,839.69).

- **HighValue flag** — A customer is flagged high-value if their balance is at or above the 75th percentile of positive balances (€139,512.29), identifying the top quartile by balance (1,596 customers, 16.0% of the base).
- **ChurnLabel / ActiveLabel** — Human-readable labels ("Retained"/"Churned" and "Active"/"Inactive") are derived from the binary Exited and IsActiveMember columns for use in chart legends.

The Surname column is dropped after loading, and any row lacking a CustomerId is discarded, ensuring the working dataframe used throughout the app is fully keyed and free of direct name-based identifiers.

4. System Architecture and Methodology

The application follows a layered, modular architecture that mirrors good practice in production data applications: a thin entry point (`app.py`) orchestrates a sequence of pure, testable functions rather than embedding business logic directly in the UI code. Figure 8 illustrates the data and control flow.

Figure 8. System Architecture of the Streamlit Analytics Dashboard

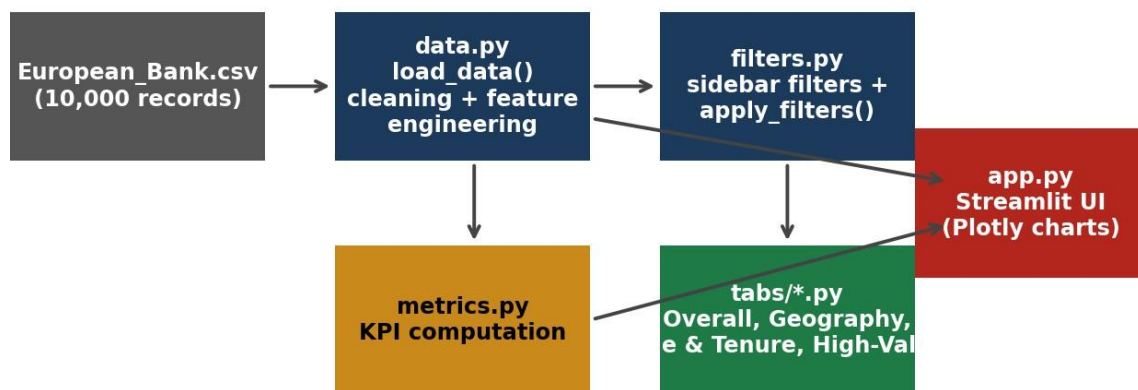


Figure 8. System architecture of the Streamlit analytics dashboard, showing the flow from the raw CSV through cleaning, filtering, KPI computation, and the four analytical tabs.

4.1 Module Responsibilities

Module	Responsibility
config.py	Streamlit page configuration (title, icon, wide layout) and the DATA_PATH constant
data.py	CSV ingestion, type coercion, missing-value handling, and all derived-feature engineering (cached via <code>@st.cache_data</code>)
filters.py	Renders the sidebar controls and applies the combined boolean filter mask to the dataframe
metrics.py	Computes and renders the five headline KPI cards shown at the top of the page
tabs/overall.py	Retained-vs-churned split, segment comparison chart, and profile comparison table
tabs/geography.py	Country-level churn rate, treemap of volume by country/churn status, and country-by-gender breakdown
tabs/age_tenure.py	Churn rate by age group, by tenure group, an age-by-tenure heatmap, and active-vs-inactive comparison
tabs/high_value.py	High-value customer KPIs, balance-vs-salary scatter plot, high-value churn-by-country chart, and a downloadable drill-down table of churned high-value customers
app.py	Application entry point: loads data, handles a missing-file fallback via file upload, renders filters, KPIs, and the four tabs

Table 2. Responsibilities of each module in the dashboard package.

4.2 Technology Stack

- **Streamlit (≥ 1.32)** — provides the reactive web UI, sidebar widgets, tabbed layout, and caching primitives.
- **Pandas (≥ 2.0) & NumPy (≥ 1.24)** — power data ingestion, cleaning, binning (`pd.cut`), and aggregation (`groupby`).
- **Plotly Express (≥ 5.18)** — renders all interactive charts: bar charts, a pie/donut chart, a treemap, a heatmap (`imshow`), and a scatter plot.

Caching the data-loading function with `@st.cache_data` ensures the CSV is parsed and the feature-engineering pipeline executed only once per session, which keeps the interface responsive as filters are toggled. The application also degrades gracefully when the CSV is not found alongside `app.py`: it surfaces a file-uploader widget so a user can supply the dataset at runtime instead of crashing.

5. Dashboard Design and Features

The dashboard is organized around a single-page layout with a persistent sidebar and a tabbed main panel, a pattern well suited to exploratory, filter-driven analysis.

5.1 Sidebar Filters

The sidebar (`filters.py`) exposes six multi-select filters — Geography, Gender, Age group, Credit score band, Tenure group, and Balance segment — plus two checkboxes for isolating active members only and

high-value customers only. All filters default to the full range of values, so the dashboard opens showing the entire customer base, and any combination of selections is applied as a conjunctive (AND) boolean mask via `apply_filters()`. A "Reset filters" button triggers a full rerun of the app to restore the defaults.

5.2 Headline KPIs

Above the tabs, five metric cards give an at-a-glance summary of the currently filtered segment: overall churn rate (with a delta versus the unfiltered baseline), high-value churn ratio, a geographic risk index (the spread in percentage points between the highest- and lowest-churn country in the current selection), an engagement-drop indicator (the gap between inactive- and active-member churn), and the active-member share of the filtered base.

5.3 Analytical Tabs

- **Overall Summary** — a retained-vs-churned donut chart, a user-selectable segment comparison bar chart (choosing among Geography, Gender, AgeGroup, CreditScoreBand, TenureGroup, or BalanceSegment), and a churned-vs-retained profile comparison table across six numeric attributes.
- **Geography** — churn rate by country, a treemap of customer volume by country and churn status, a grouped bar chart of churn by country and gender, and a supporting summary data table.
- **Age & Tenure** — churn rate by age group, churn rate by tenure group, an age-by-tenure heatmap, and an active-vs-inactive churn comparison.
- **High-Value Explorer** — KPIs for high-value customer count, churn rate, and balance at risk; a balance-vs-salary scatter plot colored by churn status and shaped by high-value flag; high-value churn rate by country; and a sortable, downloadable drill-down table of every high-value customer who churned.

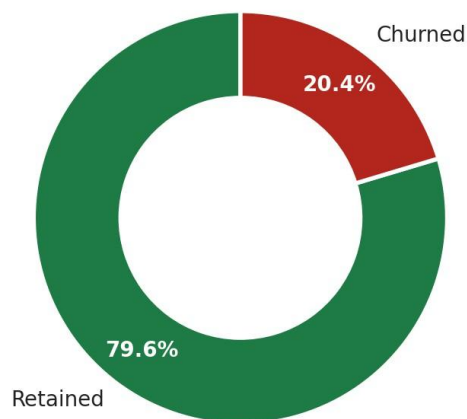
6. Empirical Findings

To validate and illustrate the dashboard's analytical value, we computed the same aggregations the application produces directly against the source CSV. All figures below reflect the complete, unfiltered dataset (n = 10,000).

6.1 Overall Churn Rate

Of the 10,000 customers in the dataset, 2,037 have churned, for an overall churn rate of **20.4%** (Figure 1). This baseline is the reference point against which every segment-level rate in the dashboard is implicitly compared.

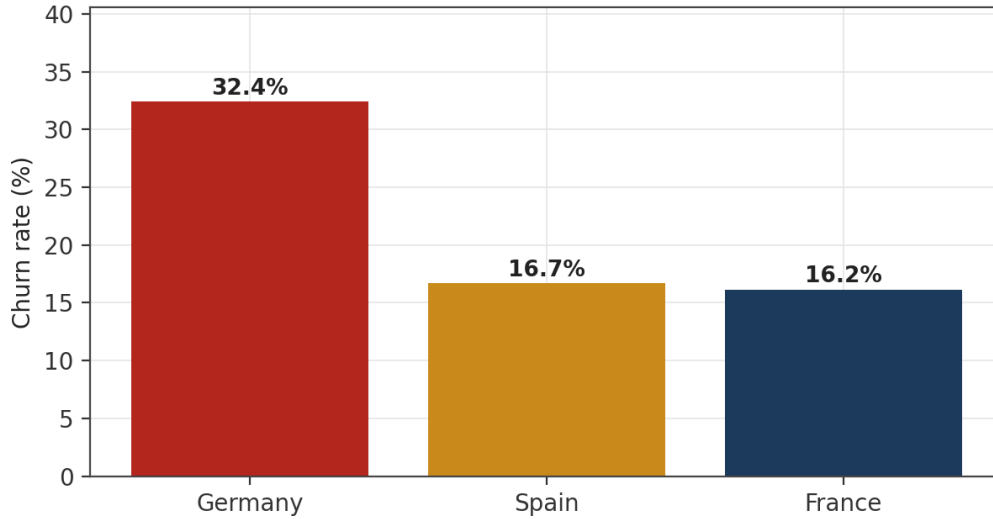
Figure 1. Overall Customer Churn Distribution (n = 10,000)



6.2 Geographic Concentration of Churn

Churn is highly concentrated by geography. Germany exhibits a churn rate of **32.4%**, roughly double the rate observed in France (16.2%) and Spain (16.7%), despite Germany holding a smaller share of the customer base (2,509 customers, 25.1%) than France (5,014 customers, 50.1%). This produces a geographic risk index of approximately 16.3 percentage points on the full dataset (Figure 2).

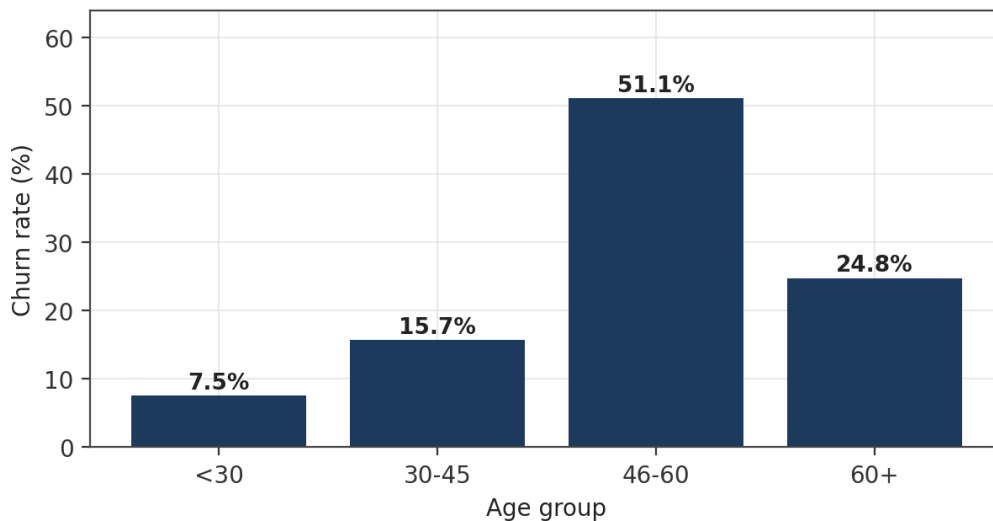
Figure 2. Churn Rate by Country of Residence



6.3 Age as a Churn Driver

Age shows a strong, non-linear relationship with churn. Customers under 30 churn at only 7.5%, while the 46-60 cohort churns at **51.1%** — more than six times the youngest group's rate — before falling back to 24.8% among customers aged 60 and above (Figure 3). The 46-60 group therefore represents the single highest-risk age segment in the portfolio, even though it is not the largest by volume (1,647 customers, versus 5,921 in the 30-45 cohort).

Figure 3. Churn Rate by Age Group

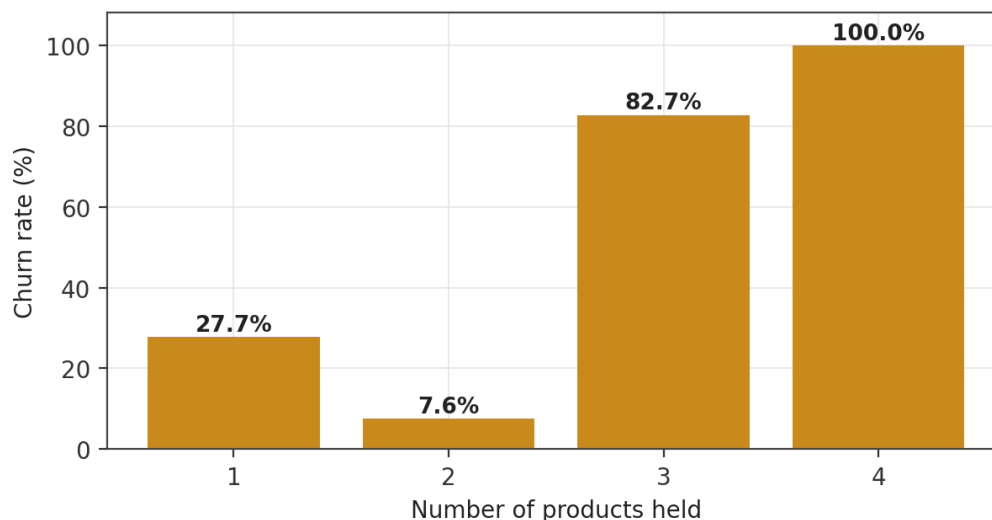


6.4 Product Engagement: A Non-Monotonic Relationship

Perhaps the most striking pattern in the dataset concerns the number of products a customer holds. Customers with exactly two products churn at only 7.6%, the lowest rate of any segment analyzed in this paper. However, churn rises sharply for customers holding three products (82.7%) and reaches 100% for

the 60 customers holding four products (Figure 4). This U-shaped, then sharply rising pattern suggests that cross-selling a second product is strongly associated with retention, while customers pushed into (or who accumulate) a third or fourth product represent an almost universally at-risk group — quite possibly customers already in the process of disengaging who are cross-sold additional products shortly before leaving, or customers with product bundles that no longer fit their needs.

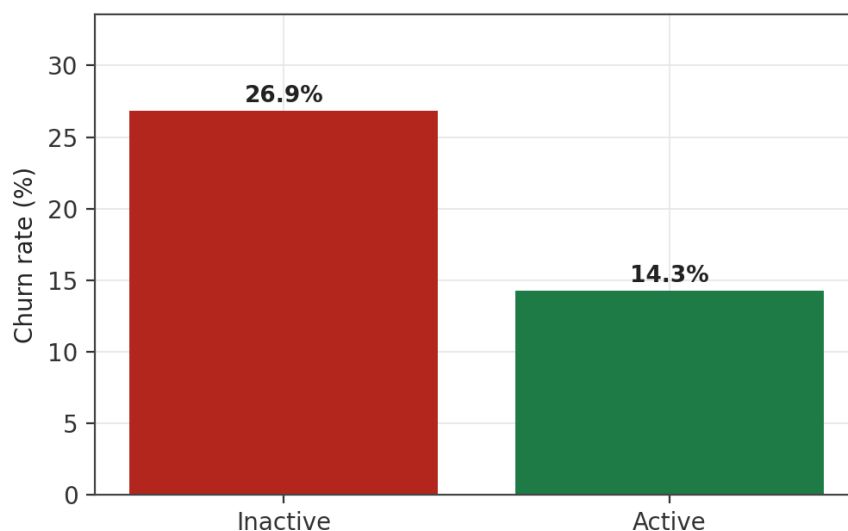
Figure 4. Churn Rate by Number of Products Held



6.5 Engagement and Activity Status

Active members churn at 14.3%, compared with 26.9% among inactive members — an engagement-drop gap of 12.6 percentage points (Figure 5). This confirms that the `IsActiveMember` flag, despite being a coarse binary indicator, carries meaningful predictive signal and is a reasonable low-cost early-warning flag for retention teams.

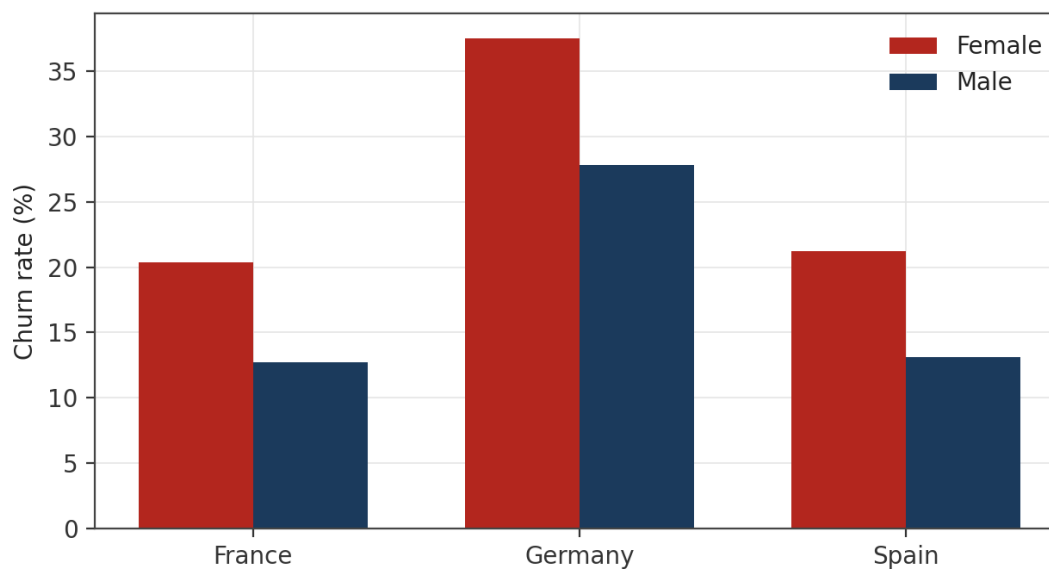
Figure 5. Churn Rate: Active vs. Inactive Members



6.6 Gender and the Country-Gender Interaction

Female customers churn at a higher rate than male customers overall (25.1% vs. 16.5%), and this gap persists within every country: in Germany specifically, female churn reaches **37.6%** versus 27.8% for men, the highest rate of any country-gender combination in the dataset (Figure 6). France and Spain show a similar, smaller gender gap of roughly 7-8 percentage points.

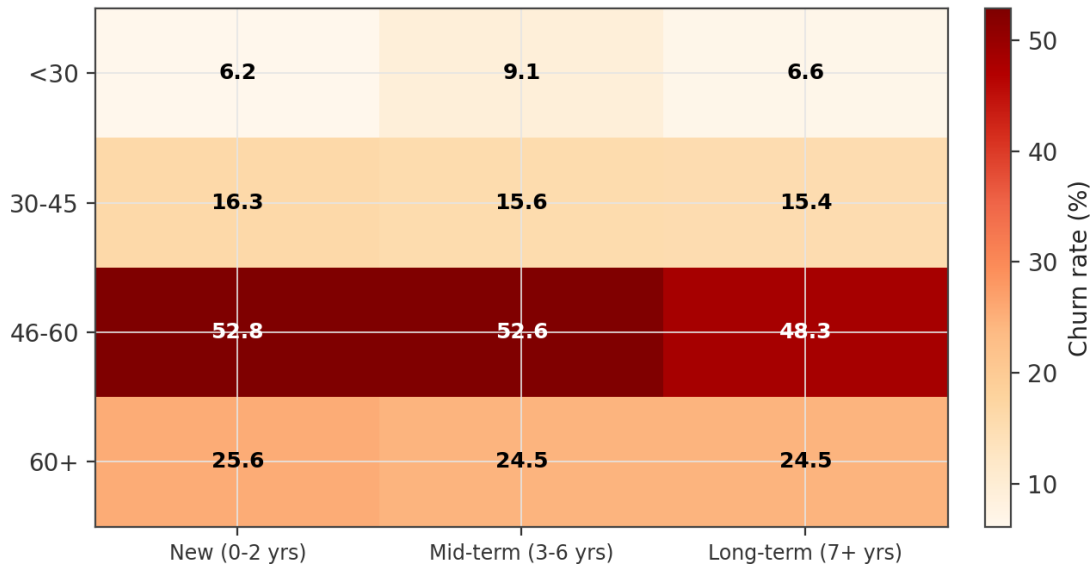
Figure 6. Churn Rate by Country and Gender



6.7 Age-by-Tenure Interaction

The age-by-tenure heatmap (Figure 7) shows that tenure alone has only a modest effect on churn (21.2% for new customers, 3-7 pp lower for longer tenure), but the interaction with age is more informative: high churn in the 46-60 age band persists across all tenure groups, indicating that age-related churn drivers (e.g., life-stage financial changes, competitor offers targeted at older affluent customers) are not simply an artifact of relationship length.

Figure 7. Churn Rate (%) — Age Group x Tenure Group



6.8 High-Value Customers

The 1,596 customers flagged as high-value (top quartile of positive account balances, threshold €139,512) churn at 23.0%, moderately higher than the 19.9% rate among the remaining base. Because these customers carry disproportionately large balances, even a moderately elevated churn rate translates into substantial balance at risk, which is precisely the reason the dashboard dedicates a full tab and a downloadable drill-down table to this segment.

6.9 Summary Table

Segment	Churn Rate	Notes
Overall baseline	20.4%	n = 10,000
Germany	32.4%	Highest-risk country
France	16.2%	Lowest, tied with Spain
Spain	16.7%	
Age 46-60	51.1%	Highest-risk age cohort
Age <30	7.5%	Lowest-risk age cohort
3 products	82.7%	Sharp risk increase vs. 2 products
2 products	7.6%	Lowest-risk product segment
4 products	100.0%	n = 60; universal churn
Inactive members	26.9%	vs. 14.3% for active members
Germany, Female	37.6%	Highest country-gender combination
High-value customers	23.0%	Top quartile by balance

Table 3. Summary of churn rates across the segments discussed in Section 6, computed directly from *European_Bank.csv*.

7. Discussion

The empirical patterns uncovered through the dashboard point to several actionable retention priorities. First, the German market warrants focused investigation: a churn rate double that of the other two countries, combined with an even higher rate among German women, suggests either a competitive dynamic specific to that market, a service or pricing mismatch, or a data-generating process that differs from France and Spain. Second, the sharp rise in churn among customers with three or more products is counter-intuitive relative to conventional cross-sell wisdom and merits root-cause analysis — it is plausible that product accumulation in this dataset reflects a reactive sales response to early churn signals rather than a cause of loyalty. Third, the concentration of churn risk in the 46-60 age band, independent of tenure, suggests that life-stage-targeted retention offers (e.g., around retirement planning or wealth consolidation) could be more effective than generic loyalty programs.

From a tooling perspective, the modular architecture described in Section 4 demonstrates that a small, well-organized Python codebase (roughly 300-400 lines across eight files) is sufficient to deliver an interactive analytics experience that would traditionally require a dedicated BI platform license. The separation of data preparation (*data.py*), filtering (*filters.py*), and presentation (the *tabs* package) means the underlying churn definitions, bin edges, and thresholds can be revised in one place without touching the visualization code, which is a meaningful maintainability advantage for a tool intended to be handed off to a business analytics team.

8. Limitations

- **Descriptive, not predictive.** The dashboard reports historical association, not forecasts; it does not estimate individual churn probabilities or support what-if scenario testing.
- **Single-year snapshot.** The dataset carries a single Year value (2025) for every record, so no time-series or cohort-over-time analysis is currently possible; churn is measured as a lifetime indicator rather than a period-over-period event.
- **Static bin edges.** Age, tenure, and credit-score bands are hard-coded in the data-loading layer; these thresholds were chosen for interpretability but are not empirically optimized (e.g., via decision-tree splits) and may obscure finer-grained patterns.
- **No statistical significance testing.** Differences between segments (e.g., the German vs. French churn gap) are reported as descriptive statistics without confidence intervals or hypothesis tests, so some smaller differences observed under heavy filtering could reflect sample-size noise.
- **Synthetic/anonymized context.** As with many publicly available banking datasets of this kind, the records are understood to be synthetic or de-identified for demonstration purposes, so findings should not be read as a claim about any real financial institution.

9. Conclusion and Future Work

This paper described a modular Streamlit dashboard for customer segmentation and churn analytics in a European retail banking context, and used it as the analytical lens through which to examine a 10,000-record customer dataset. The findings show that churn is far from uniform: geography, age, product-holding count, and engagement status each carry strong, independently discernible signal, with Germany, the 46-60 age cohort, and customers holding three-plus products standing out as the highest-priority segments for retention intervention.

Future extensions of this work could include: (1) integrating a supervised churn-prediction model (e.g., gradient-boosted trees) alongside the existing descriptive views, surfaced as an additional tab with per-customer risk scores; (2) adding cohort and time-series analysis once longitudinal data becomes available; (3) empirically optimizing bin thresholds using decision-tree-based discretization rather than fixed cut points; (4) incorporating statistical significance testing (e.g., chi-square tests for categorical segment comparisons) directly into the KPI layer; and (5) extending the high-value explorer into a lightweight customer-lifetime-value (CLV) estimator to better prioritize retention spend.

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